

Executive Memo: Collections Performance & Strategy Audit

What happened?

- The business currently reports an 11% month-on-month improvement in recovery.
- Our independent reconstruction reveals this reported 11% improvement is entirely misleading and incorrect.
- True recovery performance is highly volatile and essentially flat.
- The artificial spike was caused by data ingestion traps—specifically, exactly 500 duplicated `payment_id` retries artificially inflated the raw recovery metrics by ₹2.59 Cr.

Why did it happen?

- The metric oscillation and fake growth were driven by uncorrected data ingestion errors and operational noise rather than actual portfolio improvements.
- The underlying data pipelines suffer from severe timestamp corruption (late-arriving events) and mixed timezones across UTC, Asia/Kolkata, and Asia/Dubai.
- Legacy metrics were structurally flawed due to legacy disposition codes and schema changes.
- For example, the PTP (Promise to Pay) Rate missed exactly half of the successful promises because the system simultaneously used PTP and `PROMISE_TO_PAY` disposition codes.

How confident are we?

- We are highly confident in these findings.
- Our conclusions are derived from a strictly deduplicated "Golden Dataset" built to bypass inconsistent identifiers and overwritten historical records.
- By executing a risk-stratified portfolio mix analysis, we successfully ruled out Simpson's Paradox, confirming that the flat recovery baseline is a mathematical fact, not an anomaly hidden by shifting risk segments.

What should we do?

- Leadership has the option to invest ₹10 Cr in one of six operational areas.
- Recommendation: Place a strict HOLD on the full ₹10 Cr investment.
- Because legacy and new strategies ran concurrently in the market, we utilized a counterfactual matching model to isolate the true causal impact of the new campaigns.
- Initial unnormalized metrics falsely suggested the new "Better borrower targeting" strategy was a massive success. However, our statistical checks revealed a severe attribution-window bias: accounts in the treatment group simply had more days to accrue payments.
- The data is currently insufficient to justify a ₹10 Cr rollout. We must run a strictly controlled A/B pilot with rigid observation windows to prove actual causal lift before deploying capital.

What is the expected financial impact?

- To evaluate true ROI, we enforced a strict 90-day observation window post-intervention for both legacy and new strategy groups to neutralize the time bias.
- When normalized for time, the new targeting strategies generate a minor incremental lift of only ₹1,094.47 per account over the legacy baseline.
- We ran a 1,000-iteration bootstrap statistical test on this normalized data, which yielded a 95% Confidence Interval of [-₹329.79 to ₹2,304.51].
- Because this interval crosses zero, the observed ₹1,094 lift is statistically indistinguishable from noise. We cannot guarantee a positive ROI or forecast a reliable break-even point for the ₹10 Cr investment at this stage.